

SQL Databases $\equiv$ Relational Databases
↓
Tables.

SQL : Structured Query Language.

↓
Used to interact with
Relational DBs.

SQL : CRUD operations.

Create

Read

Update

Delete.

Read.

→ SELECT.

DROP table masai_students;

-- Primary Key : Unique and Not Null

```
create table masai_students(  
id int primary key auto_increment,  
name varchar(50) not null,  
email varchar(30) not null UNIQUE,  
address varchar(100)  
);
```

select * from masai_students;

```
insert into masai_students (name, email, address) values ('Nigam', 'nigam@gmail.com', 'India');  
insert into masai_students (name, email, address) values ('Nisarg', 'nisarg@gmail.com', 'India');  
insert into masai_students (name, email, address) values ('Abhishek', 'abhishek@gmail.com', 'India');  
insert into masai_students (name, email, address) values ('Shourya', 'Shourya@gmail.com', 'India');  
insert into masai_students (name, email) values ('Mukul', 'mukul@gmail.com');
```

delete from masai_students;

set sql_safe_updates = 1;

-- Error Code: 1175. You are using safe update mode and you tried to update a table without a WHERE that uses a KEY column. To disable safe mode, toggle the option in Preferences -> SQL Editor and reconnect.

```
insert into masai_students (name, email, address) values  
('dfjklsd', 'sdlkfjlk@gmail.com', 'India');
```

-- Error Code: 1062. Duplicate entry 'nigam@gmail.com' for key 'masai_students.email'
-- Error Code: 1062. Duplicate entry 'Nisarg@gmail.com' for key 'masai_students.email'
-- Error Code: 1062. Duplicate entry 'abhiShek@gmail.com' for key 'masai_students.email'
-- Error Code: 1406. Data too long for column 'name' at row 1

-- ORDER BY

-- default order of sorting - ascending.

-- sort in descending order

select * from film order by length desc limit 20;

-- from students table, get top 10 students based on the percentage.

-- select * from students order by percentage desc limit 10;

-- 10 recent orders

-- select * from orders order by created_at desc limit 10;

-- CRUD - Create, Read, Update

```
select * from film;
```

```
update film set length = 630 where film_id = 10;
```

-- You are using safe update mode and you tried to update a table without a WHERE that uses a KEY column. To disable safe mode, toggle the option in Preferences -> SQL Editor and reconnect.

-- ALTER : update/alter the table (add or remove a column, add/remove a constraint,)

```
ALTER table masai_students  
modify address varchar(100) not null;
```

```
ALTER table masai_students  
modify address varchar(100) unique;
```

-- Error Code: 1138. Invalid use of NULL value

-- Error Code: 1364. Field 'address' doesn't have a default value

-- Create, Read, Update, Delete

